INFO 202: Data Curation

Lanzhou University

Week 1 – Session 3

Lecturer: Orlando Ribas Fernandes

Course Plan – Week 1 *(updated)*

Week	Topics		Tasks (Practice)
Week 1 Session 1 20-Apr 8:30 / 14:30	• Intro	online	

Week 1
 	
Session 2
 	
23-Apr
8:30 / 10:30
	•Intro to Data Curation
	    • What data curation really is (vs scraping / analytics)
	    • Why traditional trend reports fail
	    • Industry examples
	
	•Data types: text, image, video, context
	• Structured vs unstructured data
	• Images matter?
		online	Mini-case analysis:
			How two brands interpreted the same trend differently
			
			Distinguish signal vs noise

Week 1 Session 3 24-Apr 14:30 / 16:30	• **Project** Introduction • Full explanation of the course project challenge	online	Group formation Initial domain + data ideas

The Project (Challenges + What We Expect)

Trend Intelligence for Brands (AI-Assisted Data Curation)

How can a brand use public data sources to detect trends in a defined period and turn them into recommendations (designs, colours, shapes, patterns, themes)?

- Current challenges
  - Data is fragmented: trend signals are spread across many platforms and formats
  - Data is noisy and biased: popularity ≠ relevance; platforms distort what we see
  - Multimodal complexity: images contain signals that text alone cannot capture
  - Lack of structure: without metadata and consistent labels, insights are not reusable
  - Ethics & privacy: public data still requires responsible use and documentation
  - Automation trade-offs: AI can help, but human judgement remains essential

The Project (Challenges + What We Expect) (2)

- What we expect
  - A curated dataset built from multiple datasources (social media influencers, brands, ecommerce, websites, pdfs, videos, etc..)
  - A documented curation pipeline (collection → cleaning → metadata → versioning)
  - Trend insights grounded in evidence:
    - themes/topics + visual patterns
    - colours, materials, shapes/silhouettes, patterns, and emerging signals
  - Actionable brand recommendations
    - product/creative directions aligned with trends
    - opportunities for differentiation (not just “what is popular”)
  - An AI-assisted component
    - a simple AI agent concept or prototype to support scraping/validation/curation
  - Ethics & risk statement
    - bias considerations, privacy constraints, and responsible practices

The Project (Challenges + What We Expect) (3)

- Milestones
  - Week 3 – Presentation 1: problem statement + data sources + approach + implementation plan
  - Week 6 – Presentation 2: progress update + what changed + early results + next steps
  - Week 9 – Final: validated results + insights + recommendations + demo/storytelling

- Bonus
  - Bonus 1: Full agents implementation that help automate repetitive work (scrape → validate → log → label), while keeping human oversight for interpretation.
  - Bonus 2: Cloud of Words for Images: Instead of just text, students will use images, shapes and patterns from Instagram/TikTok images, etc.. then create a new Visual representations.

Presentation 1

- Milestones
  - Week 3 – Presentation 1: problem statement + data sources + approach + implementation plan
- Deliverables:
  - Week 3 – Complete report about the project (in PDF) + Powerpoint presentation (in PDF)
- Dates:
  - Zip File submission: 8-May (deadline)
  - Presentation: 9-May – in class, where each group will have 5 min

Grades

Task	Weight	Date
Course participation	10%	
Assignment 1 – Project problem & plan presentation	20%	Week 3
Assignment 2 – Project first results	30%	Week 6
Assignment 3 – Final Project Presentation	40%	Week 9

Rules

- The projects must be proposed and developed by student groups each of which consists of 4 students.

- The total marks are 100.
- Work in group of 4.
- University Academic Honesty Rules and Procedures will be adhered to strictly.
- Zero tolerance will be given to academic fraud of any sort, including copying content from the internet and/or other assignments.
- Please name your assignment file as INFO202-<finalproject>-group-<number>.zip, with the required deliverables.
- Submit your assignment file via moodle until the due data. Email if the moodle is offline.
- Assignments will have heavy penalties beyond that point. Missing work will earn a zero grade.
- Images must be clear and legible. Assignments will be judged on the basis of code, visual appearance, grammatical correctness, and quality of writing, as well as their contents. Please make sure that the text of your assignments is well-structured, using paragraphs, full sentences, and other features of well-written presentation. Text font size should be at least 11 points.
- Presentation will follow the group numbers order.
- All text shall be in English.
- Details about the assignments, rules on Moodle

Projects - Some tips:

- Start early and manage your time effectively.
- Organize the tasks between the team members
- Keep your data and code organized and well-commented.
- Follow the best practices – documentation, code and presentations
- Practice your presentation multiple times.
- Be prepared to answer questions about your work.

General Best pratices

Know Your Audience

- Identify your audience
- Understand their needs
- Tailor your proposal accordingly

Research and Planning

- Conduct thorough research
- Plan your project
- Identify key resources

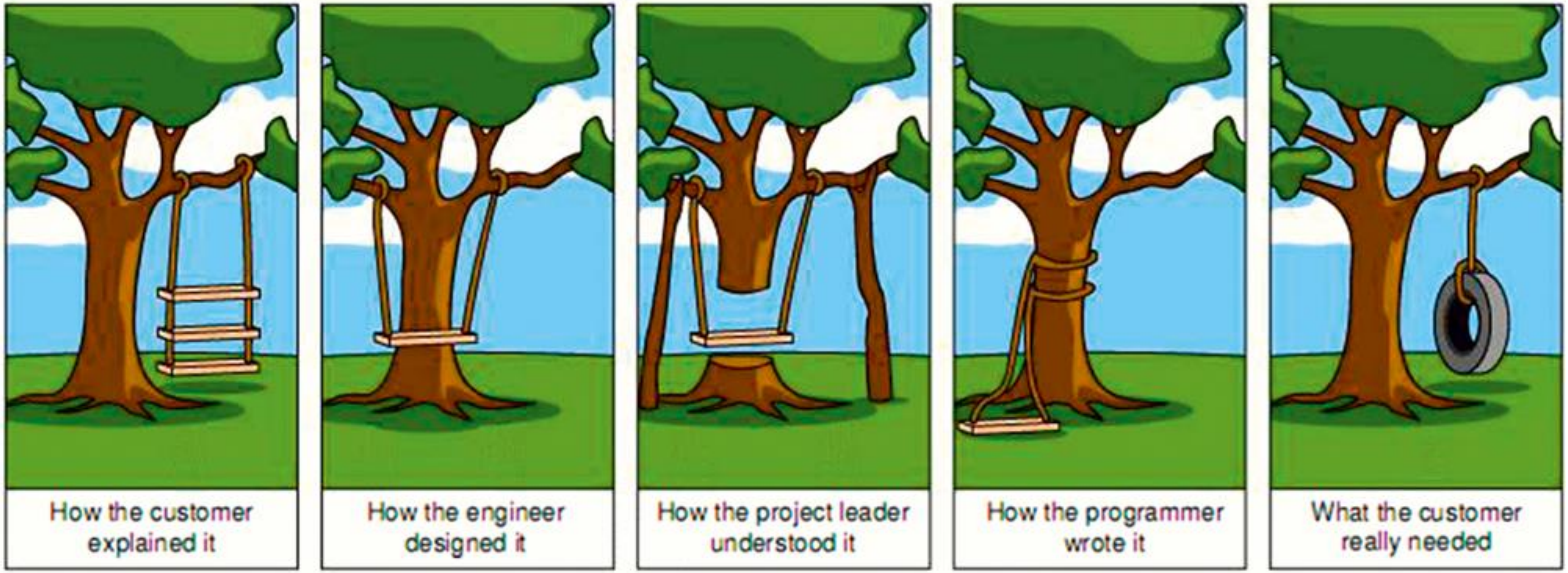
Project Objectives and Goals

- Define project objectives
- Set achievable goals
- Align project goals with organizational objectives

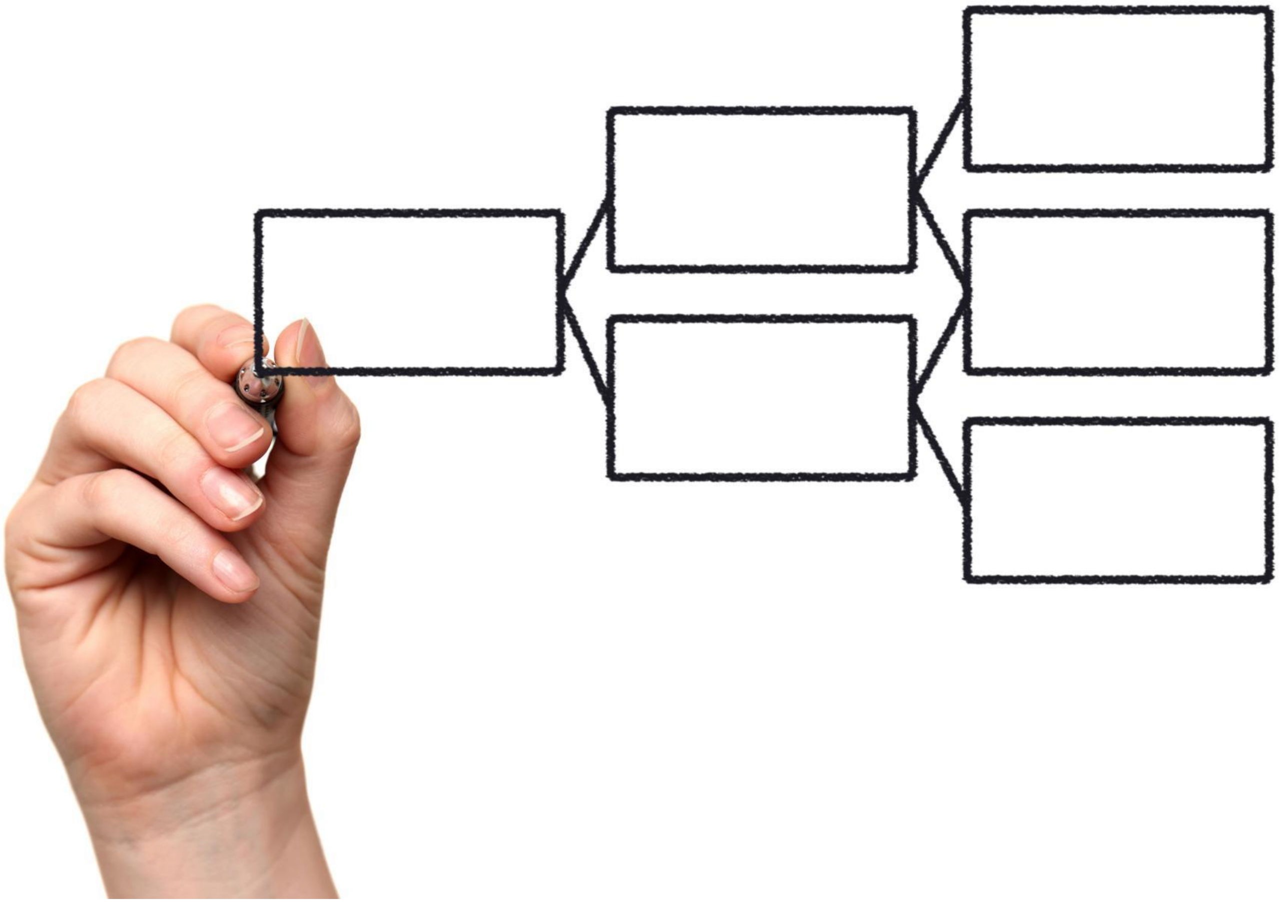
Project Scope

- Define project scope
- Identify project boundaries
- Ensure feasibility

Project Scope



Methodology



- Describe your methodology
- Explain how you will collect and analyze data
- Establish a project timeline

Risk Management

- Identify potential risks
- Develop a risk management plan
- Monitor and manage risks

Conclusion

- Summary of key takeaways

Project Documents

- Report:
  - Following the best practices.
  - Professional format
  - Cover page
  - Introductions and conclusions
  - All the required topics to “sell” the project.

- Presentation:
  - 5 min each group.
  - Limited time.
  - Prepare in order to pass the message and “sell” the project.

Reasons for deducing points
(from previous years)

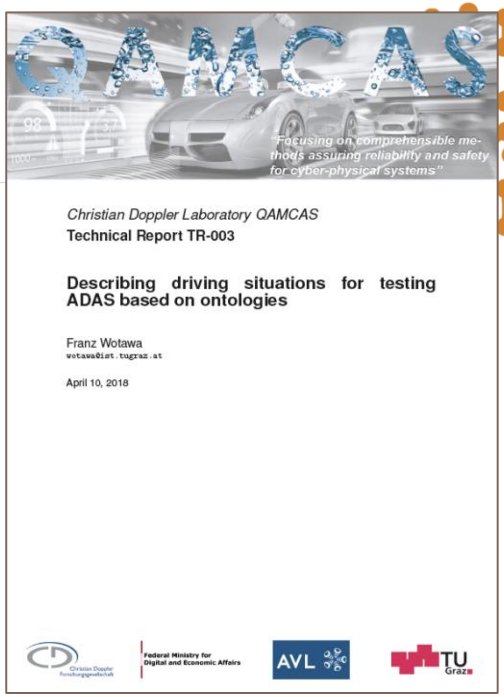
- **Minimalistic solutions** (mostly coupled with a not-so-good use of English)
- **Please follow the guidelines for writing a report!**
  - **TITLE PAGE!!!!**
- Use a good name for your report file (including the assignment number and your group name)!
- ZIP files are only allowed to be submitted via Moodle for assignment

Writing a report for an assignment

Document structure

- Title page
- Table of contents
- Introduction
- *Structure representing*
- *the content/objectives/…*
- Conclusions
- References (if required)
- Appendix (if required)

Please add all team members (because they are authors) AND the group number or name to the title page!!!!


QAMCAS

"Focusing on comprehensible methods assuring reliability and safety for cyber-physical systems"

Christian Doppler Laboratory QAMCAS
Technical Report TR-003

Describing driving situations for testing ADAS based on ontologies

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April 10, 2018

In all these parts we describe the content of the report. We always start with and introduction and end the report with a conclusion (or summary). In between we structure the document so that it reflects the content.

Document structure - Example

Reasoning from first principles for self-adaptive and autonomous systems

Franz Wotawa

Abstract Model-based reasoning or reasoning from first principle is a well known method for performing various tasks including diagnosis from systems' models directly. In this chapter, will first discuss the basic principles and algorithms of model-based reasoning relying on the system models of the correct behavior as well as fault models. Afterwards, we discuss how to provide models including a discussion of the use abstraction. We further on extend the basic foundations allowing model-based diagnosis to be applied to self-adaptive systems including fail-operational system. Beside the system architecture comprising monitoring capabilities, we show how to integrate a model-based diagnosis engine enabling the system to reason about its internal fault state and to take appropriate actions for repairing and compensating detected and localized faults. We illustrate the underlying concepts using an autonomous mobile robot as example where we focus on the robot's drive.

1 Introduction

Systems that are able to automatically adapt their behavior in cases of faults have always been very much appealing for research and practice. Because of the increasing importance of applications like autonomous vehicles, the internet of things (IoT), or industry 4.0 dealing with increased autonomy, self-adaptation increases importance as well. For example, consider a truly autonomous vehicle transporting passengers from one location to another. If there is a system fault occurring during operation, there is no human driver working as fall-back mechanism for assuring that the vehicle goes to a safe state, e.g., driving to an emergency lane of a highway and stopping there. In case of autonomous driving the system itself is responsible for any action

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for self-adaptive systems. Therefore, we also discuss related research and previous work that has been published.

This chapter is organized as follows: We first introduce the application domain of autonomous mobile robots in Section 2. Afterwards, in Section 3 we introduce the basic concepts of model based reasoning including model-based diagnosis and abductive diagnosis. In Section 4 we discuss issues of modeling for model-based reasoning, followed by Section 5 where we introduce the basic architectures behind a self-adaptive system. There we make use of three examples outlining the different repair actions necessary to bring the system back into an operational state. Finally, we discuss related literature in Section 6 and conclude the paper in Section 7.

2 Example

In this section, we introduce the example of a mobile robot having a differential drive for moving from one point in a plain to another. We will use this example in our chapter for introducing the basic concepts behind model-based reasoning and in an extended form for showing how self-adaptive behavior can be implemented using models of the system directly.

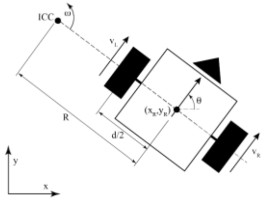

Fig. 1 Mobile robot with differential drive.

A differential drive comprises two wheels with varying speed. Depending on the speed of the wheels the robot either rotates, moves on a straight line, or on a curve. In the following, we discuss a kinematics model of a mobile robot with a differential drive. For more details we refer the interested reader to [13]. In Figure 1 we show the underlying ingredients. We assume that the robot is at its current position (x_R, y_R)

show how such abstract models can be used for identifying the cause of a detected misbehavior. For example, we will consider the case of a differential drive robot that follows a wrong trajectory. In Figure 2 we depict such a case, where a robot is expected to rotate clockwise followed by moving on a straight line but follows a curve. Obviously in this case either the speed of the left wheel is too low or the one of the right wheel too high.

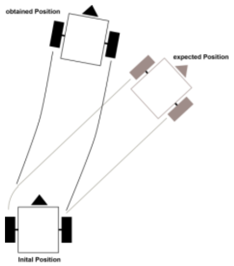

Fig. 2 A small mobile robot driving the wrong trajectory.

3 Model-based reasoning

The underlying idea behind model-based reasoning is to use a model of a system directly to reason about the system. In one instance, i.e., model-based diagnosis, the system's model is used for identifying root causes in case of an observed behavior that contradicts the expected one. A model in model-based diagnosis (MBD) comprises the system's structure including its components and interconnections, as well as the component models. The health state of components, i.e., a predicate indicating whether a component is working as expected or not, is used to indicate a root cause. In this terminology an incorrectly working component maybe an explanation for the detected unexpected deviation in the observed behavior. Davis [11] was one of the first outlying the basic principles behind MBD that Reiter [46] and De Kleer and Williams [30] further formalized and extended.

It is worth noting that in classical MBD the component models only describe the correct behavior. This makes the theory general applicable even in cases where there is no knowledge about faults and their consequences available. De Kleer et

model, created at design time as Labelled Transition System (LTS) that is used for checking properties.

In the context of cyber-physical systems there has also been research published. Niggemann and Lohweg [42] discussed the state of the art of diagnosis of cyber-physical systems and the research questions but focussing more on production systems. Mahadevan and colleagues [34] introduced the application of causal diagrams for fault localization. The approach presented in this work relies in contrast to these papers on model-based reasoning based on formal models.

7 Conclusions

In this chapter, we discussed the basic principles and foundations behind model-based reasoning in detail. We motivated why reasoning based on models is of particular interest for self-adaptive systems using a running example from the autonomous mobile robot domain. We illustrated different challenges that occur in this domain and discussed possible solutions. Regarding diagnosis we introduced two different methods, i.e., (1) model-based diagnosis, and (2) abductive diagnosis. The first relies on models of system components only considering the correct behavior. The latter requires knowledge about faults and their corresponding behavior. For both approaches we outlined a simple algorithm allowing for computing minimal diagnoses from models and observations.

Because modeling is the important part, when introducing model-based reasoning, we also explained how to model for diagnosis. We outlined a process that starts with a system architecture comprising components and connections. From this discussed how to come up with abstract models that maps potentially infinite domains into a small set. For this purpose, we referred to qualitative reasoning and the abstractions discussed there. Furthermore, we introduced concepts for extracting models from other development artifacts like FMEAs or simulation models.

After presenting the basic concepts, we focused on the integration of diagnosis for self-adaptive systems. There we introduced a system combining a smart diagnostics with an ordinary control system interacting via sensors and actuators with its environment. We discussed the basic repair cycle and an algorithm for combining diagnosis with repair. In contrast to previous work we did not only rely on simple repair actions, e.g., replacing one component with its spare part, but also discussed how compensating actions can be integrated and used in the proposed setting. For the latter it is worth noting that the system model itself can be used to obtain sufficient information about the compensation.

The presented approach can be integrated into cyber-physical systems like autonomous vehicles in order to implement fail-operational behavior where compensating repair as well as automated repair using redundant hardware that can be enabled during operation is required.

Conclusion